

CEN 201 Object Oriented Programming Midterm Exam

Student Name:

25-11-2020

Student #:

Daytime : ☐

Evening: ☐

(30)1- Explain the “this” pointer in details.

(30)2- a) Why do we use constructor?

b) Why do we use destructor?

(50)3- Assume you have the following program. When you execute the program, the computer will generate error because you are passing an object to a function by value. You need to write a copy constructor to solve the problem. Write the copy constructor for the given class.

```
#include <iostream>
using namespace std;
class myclass {int size;int *p;
public:
double x,y,z;
myclass (int sz) { //constructor
p=new int[sz];
if (!p) exit(1);
size=sz;
for (i=0;i<size;i++) p[i]=2*i;
}
~myclass() {delete [] p;} //destructor
int get(int i) {return p[i];}
};
void show(myclass a)
{int i;
cout<< "x= " <<a.x<<endl;
cout<< "y= " <<a.y<<endl;
cout<< "z= " <<a.z<<endl;
cout<<"array : "
for (i=0;i<a.size;i++) cout<< a.get(i); cout<<" ";
cout<< endl;}
```



```
int main () { myclass obj1(10);
show(obj1);
return 0;}
```